Lecture 28.13: Variance Reduction as Impurity Metric for Regression Trees

📌 Executive Summary

Mathematical formulation of Variance Reduction, Mean Squared Error (MSE) node impurity, Mean Absolute Error (MAE) median criteria, and split evaluation for regression trees.

🏛️ Theoretical Formulation & Mathematical Architecture

📉 Variance Reduction Formulation

In regression trees, node impurity is defined by the **Sample Variance** (or Mean Squared Error) of the continuous supervisory target y within node S:

$$I_{\text{MSE}}(S) = \frac{1}{|S|} \sum_{i \in S} (y_i - \bar{y}_S)^2 = \text{Var}(y \mid i \in S)$$

For a proposed split on feature A generating left subset S_L and right subset S_R:

$$\boxed{\Delta V(S, A) = I_{\text{MSE}}(S) - \left(\frac{|S_L|}{|S|} I_{\text{MSE}}(S_L) + \frac{|S_R|}{|S|} I_{\text{MSE}}(S_R) \right)}$$

The optimal split maximizes the reduction in variance $\Delta V(S, A)$, forcing children to be tightly clustered around their respective local means.

🛡️ Mean Absolute Error (MAE) Median Alternative

Under L_1 loss:

$$J_{\text{MAE}}(c) = \sum_{i \in S} |y_i - c|$$

The minimizing constant is the **sample median**:

$$c^* = \text{median}(y_i \mid i \in S)$$

Impurity under MAE is the Mean Absolute Deviation from the median:

$$I_{\text{MAE}}(S) = \frac{1}{|S|} \sum_{i \in S} |y_i - \text{median}(y)|$$

MAE provides extreme resistance against wild regression outliers (such as ₹10 crore luxury property transactions in standard housing data).

💻 Vectorized Python Implementation

```
import numpy as np

def compute_variance_reduction(parent_y, left_y, right_y):
    n_p, n_l, n_r = len(parent_y), len(left_y), len(right_y)
    var_p = np.var(parent_y)
    weighted_child_var = (n_l / n_p) * np.var(left_y) + (n_r / n_p) * np.var(right
    return var_p - weighted_child_var

# Test
p_vals = np.array([1.0, 2.0, 9.0, 10.0])
l_vals = np.array([1.0, 2.0])       # tight cluster around 1.5
r_vals = np.array([9.0, 10.0])      # tight cluster around 9.5

vr = compute_variance_reduction(p_vals, l_vals, r_vals)
print("Variance Reduction for clean split:", round(vr, 4))
assert vr > 0, "Clean separation must reduce continuous variance"
```

🔬 Key Engineering Takeaways & Invariants

1. **Axis-Aligned Partitioning**: Decision trees recursively split feature space along orthogonal boundaries without requiring feature scaling.
2. **Impurity Optimization**: Split selection deterministically maximizes impurity reduction (Information Gain for classification, Variance Reduction for regression).

3. **Generalization Safeguards**: Regularization via pre-pruning (`max_depth` , `min_samples_leaf`) and post-pruning (`ccp_alpha`) prevents noisy leaf memorization.